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Detailed Course

Introduction and Motivation

General Context

PCA (Principal Component Analysis) is a **linear latent decomposition** method that identifies the **underlying latent factors** structuring data.

Main Problem:

Base representations of data are not always optimal. PCA seeks to find a new representation that:

- Reveals the **latent factors** (hidden) governing the process
- Uses **statistical correlation** to identify these factors
- Enables **simplification** of data through justified decimation

Fundamental Characteristics

PCA exhibits **linear latent factors** and allows several complementary interpretations that we will study.

Key Point: PCA belongs to the family of **linear latent models**

Formalization

Maximizing Variance

Geometric Intuition

Fundamental Question: What information should we preserve when projecting?

Illustrative Example:

Suppose data $X \subset \Omega$ is almost contained within a 2D plane in a 3D space. The relevant choice is to select a basis $\{u_1, u_2\}$ for this plane.

Characterization of the Optimal Basis $\{u_1, u_2, u_3\}$:

- Data **varies most** along directions u_1 and u_2
- Data **varies least** orthogonally to these directions (along u_3)

Mathematical Formalization

Definitions:

Let X be the dataset, $\{u_i\}_{i \in [\![D]\!]}$ a new **orthonormal basis** of $\mathbb{R}^D$.

The **first principal component** u_1 is chosen such that the **variance of the projected data** on u_1 is maximized.

The component u_2 is then chosen using the projection on $\mathrm{Proj}_{u_1^\perp}(X)$ (orthogonal to u_1).

Mathematical Model:

The empirical mean and projected variance are:

$$x = \frac{1}{N}\sum_{i=1}^{N} x_i$$

$$v_{u_1} = \frac{1}{N}\sum_{i=1}^{N}(u_1^T x_i - u_1^T x)^2 = u_1^T \Sigma u_1$$

where $\Sigma = \frac{1}{N}\sum_{i=1}^{N}(x_i - x)(x_i - x)^T$ is the **covariance matrix** of the data.

Optimization Problem:

$$u_1 = \arg\max_{u^T u = 1} u^T \Sigma u$$

Solution via Lagrange Multipliers

Lagrangian Function:

$$J(u) = u^T \Sigma u + \lambda(1 - u^T u)$$

Optimality Conditions:

$$\left.\frac{\partial J(u)}{\partial u}\right|_{u=u_1} = 0 \quad \text{and} \quad \left.\frac{\partial J(u)}{\partial \lambda}\right|_{\lambda=\lambda_1} = 0$$

Fundamental Result:

$$\Sigma u_1 = \lambda_1 u_1 \quad \Rightarrow \quad v_{u_1} = u_1^T \Sigma u_1 = \lambda_1$$

Conclusion: (u_1, λ_1) is an **eigenpair** of the covariance matrix Σ

Complete Decomposition

Continuing this decimation process, we obtain the set of **principal components** as the eigenpairs $\{(u_i, \lambda_i)\}_{i \in [\![D]\!]}$ of Σ.

Matrix Decomposition:

$$\Lambda = \begin{pmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_D \end{pmatrix} = \begin{pmatrix} | & | & & | \\ u_1 & u_2 & \cdots & u_D \\ | & | & & | \end{pmatrix}^T \Sigma \begin{pmatrix} | & | & & | \\ u_1 & u_2 & \cdots & u_D \\ | & | & & | \end{pmatrix} = U^T \Sigma U$$

Minimizing Error

Variance-Error Equivalence

The variance v_{u_1} can be expressed as a sum of squares:

$$v_{u_1} = \frac{1}{N} \sum_{i=1}^{N} (u_1^T (x_i - x))^2$$

Reasoning via the Pythagorean Theorem:

- To maximize v_{u_1}, the terms $u_1^T(x_i - x)$ must be collectively maximized
- Since $(x_i - x)$ is fixed, this is equivalent to **minimizing the distance to the projection axis** (approximation error)

Alternative Formulation:

$$u_1 = \underset{u^T u = 1}{\arg\min} \sum_{i=1}^{N} \|(x_i - x) - [u^T (x_i - x)] u\|_2^2 \quad \Rightarrow \quad \Sigma u_1 = \lambda_1 u_1$$

Fundamental Equivalence:

Maximizing projection variance Minimizing approximation error

Interpretation: Quadratic Regression

A principal component is a **quadratic regression** on the data

Physical Interpretation

Consider a physical system X with masses $m_i = 1$ at positions x_i.

System Inertia:

The inertia with respect to a point $a \in \Omega$ is:

$$I_a(X) = \sum_{i=1}^{N} d^2(a, x_i)$$

Huygens' Theorem:

If $g = \frac{1}{N} \sum_{i=1}^{N} x_i$ (center of gravity), then:

$$I_a(X) = d^2(a, g) + I_g(X)$$

This relation is the physical analog of $\mathrm{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$.

For a subspace F passing through g:

$$I_F(X) = \sum_{i=1}^{N} d^2(F, x_i) \quad \text{where} \quad d(F, x_i) = \|x_i - \mathrm{Proj}_F(x_i)\|$$

Physical Conclusion: A principal component is a **subspace of minimal inertia** with respect to X

Dimensionality Reduction

Structure of the Latent Space

Properties of the Latent Space

The latent space with basis $\{u_1, \dots, u_D\}$ has the following properties:

Eigenvalue Order:

By construction: $\lambda_1 > \lambda_2 > \cdots > \lambda_D$

Total Variance:

$$\lambda \stackrel{\text{def}}{=} \sum_{d=1}^{D} \lambda_d = \mathrm{Tr}(\Sigma)$$

λ represents the **total variance** of the data.

Decorrelation of Latent Features:

$$v_{u_d} = \lambda_d \quad \text{and} \quad u_d^T u_{d'} = 0 \text{ for } d \neq d'$$

Λ is the **diagonal covariance matrix** in the latent space.

Latent Coordinates

The basis $\{u_1, \dots, u_D\}$ induces **latent coordinates** y_i for the data:

$$y_i(d) = \langle u_d, x_i - x \rangle = u_d^T (x_i - x)$$

In vector form:

$$y_i = U^T (x_i - x)$$

PCA Linear Transformation

The PCA transformation is **linear** and consists of three steps:

1. **Centering** on x
2. **Rotation** using matrix U
3. **Scaling** (whitening) using $\Lambda^{1/2}$

Underlying Model

Variance Preservation:

The latent space preserves the variance of the centered data.

Distribution Assumption:

$$X_i \sim f_X = \mathcal{N}(x, \Sigma)$$

Important Limitation: PCA will **not be relevant for non-Gaussian data** (e.g., clustered data)

Exact Transformation

Crucial Point: So far, PCA is an **exact transformation**

$$y_i = U^T(x_i - x) \quad \text{thus} \quad Uy_i = UU^T(x_i - x) = x_i - x$$

Since U is orthogonal: $UU^T = I_D$

At this stage, the goal is to study the **spectrum** $\{\lambda_1 > \lambda_2 > \cdots > \lambda_D\}$ of the data to understand how variance is distributed.

Low-Rank Approximation and Dimensionality Reduction

Eckart-Young Theorem

Fundamental Theorem:

If $\Sigma = U\Lambda U^T$ and for $K < D$, we define:

$$\Sigma_K \stackrel{\text{def}}{=} \sum_{d=1}^{K} \lambda_d u_d u_d^T$$

Then:

$$\underset{\text{rank}(S)=K}{\arg\min} \|\Sigma - S\|_F^2 = \Sigma_K$$

where $\|\cdot\|_F$ is the **Frobenius norm**.

Approximation Error:

$$\|\Sigma - \Sigma_K\|_F^2 = \sum_{d=K+1}^{D} \lambda_d$$

Σ_K is the **closest rank-K matrix** to Σ

Practical Truncation

Truncation of U and Λ:

$$\Sigma_K = \begin{pmatrix} | & & | \\ u_1 & \cdots & u_K \\ | & & | \end{pmatrix} \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_K \end{pmatrix} \begin{pmatrix} | & & | \\ u_1 & \cdots & u_K \\ | & & | \end{pmatrix}^T = U_K \Lambda_K U_K^T$$

Projection and Reconstruction:

- **Projection** (encoding): $\tilde{y}_i = U_K^T x_i \in \mathbb{R}^K$
- **Reconstruction** (decoding): $\tilde{x}_i = U_K U_K^T x_i + x$

Important: We go from D dimensions to K dimensions with $K < D$

Geometry and Reconstruction Quality

Relative Contribution to Variance

Contribution of Each Dimension:

$$c_d = \frac{\lambda_d}{\sum_k \lambda_k}$$

This metric depends on the **spectrum distribution** of eigenvalues.

Projection Ratio

Projection ratio of a data point x_i on a latent factor:

$$\rho_d(x_i) = \frac{\langle u_d, x_i \rangle^2}{\|x_i\|^2} = \cos^2(\angle(u_d, x_i))$$

Interpretation:

- The closer $\rho_d(x_i)$ is to 1, the more x_i lies on the axis u_d
- These metrics can be **grouped** (summed) to evaluate quality with respect to a subspace $\{u_1, u_2, ...\}$

Normalization and Scaling

Problem of Different Scales:

Each original dimension has its own unit (scale), estimated by σ_d^2 (empirical variance along dimension d).

PCA is more effective if **all scales are similar**

Solution: Normalization

We create the **scaling matrix**:

$$S = \text{diag}[\sigma_1^2, \dots, \sigma_D^2]$$

We define the **Mahalanobis metric**:

$$d_S^2(x, y) = (x - y)^T S^{-1} (x - y)$$

In this metric space, the covariance matrix Σ_S is also normalized (it becomes the **correlation matrix**) and used as the basis for PCA.

Practical Tip: Normalizing data before PCA often improves results

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Practical Application of PCA

Complete PCA Algorithm

Application Steps:

Given dataset $X \in \Omega$:

1. **Compute** the empirical mean $x = \frac{1}{N} \sum_{i=1}^{N} x_i$
2. **Center** the data: $x_i \leftarrow (x_i - x)$ and form the centered matrix X
3. **Compute** the covariance matrix: $\Sigma = \frac{1}{N} X X^T$
4. **Decompose** into eigenvalues: $\Sigma = U \Lambda U^T$

So far: **exact** transformation

5. **Select** the number of components K
6. **Define** U_K, Λ_K and compute $\{\tilde{y}_i\}_{i \in [\![N]\!]}$ and/or $\{\tilde{x}_i\}_{i \in [\![N]\!]}$

Choosing the Number of Components K

Three Main Strategies:

Strategy 1: Target Dimension

$$K = d$$

where d is the desired dimension, giving $\tilde{y}_i \in \mathbb{R}^d$

Strategy 2: Variance Threshold

Require that the latent space retains a proportion τ of the total variance:

$$\text{Var}(Y) = \tau \cdot \text{Var}(X) \quad \Rightarrow \quad K \text{ such that } \frac{\sum_{d=1}^{K} \lambda_d}{\sum_{d=1}^{D} \lambda_d} \geq \tau$$

Example: $\tau = 0.95$ means retaining 95% of the variance

Strategy 3: Reconstruction Error

Train K such that $L(X) \leq \epsilon$, where for example:

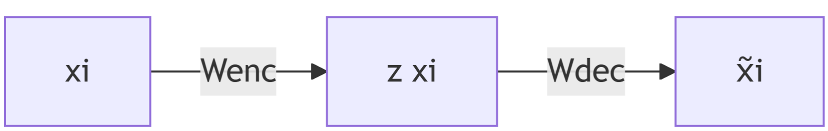
$$L(X) = \sum_i \|x_i - \tilde{x}_i\|^2$$

Practical Advice: Visualizing the eigenvalue spectrum (scree plot) helps identify a natural "elbow"

Alternatives and Extensions

Linear Autoencoders (Linear AE)

Structure of a Linear Autoencoder:



- W_{enc} and W_{dec}: weight matrices (encoder and decoder)
- $z(x_i) = W_{\text{enc}} x_i$: encoding

- $\tilde{x}_i = W_{\text{dec}} z(x_i)$: decoding

Optimization Problem:

$$\theta^* = \arg\min_{W_1, W_2} \sum_{i=1}^{N} \frac{1}{2} \sum_{d=1}^{D} (x_i[d] - \tilde{x}_i[d])^2 = \arg\min_{\text{rank}(W)=K} \|X - WZ\|_F^2$$

Relationship Between PCA and Linear Autoencoder

SVD Decomposition:

Using the Eckart-Young theorem with $X = U\Psi V^T$, the solution is:

$$W_{\text{dec}} Z = U_K \Psi_K V_K^T$$

Optimal Solution:

Setting $W_{\text{dec}} = U_K \Psi_K$, then:

$$W_{\text{enc}} = \Psi_K^{-1} U_K^T$$

Reconstruction:

$$\tilde{x}_i = U_K U_K^T x_i$$

Mathematical Relationship:

For **centered** data:

- **PCA:** $\Sigma = \frac{1}{N} X X^T = U \Lambda U^T$ thus $\tilde{x}_i = U_K U_K^T x_i$
- **AE:** $X = U\Psi V^T$ thus $\Sigma = \frac{1}{N} U \Psi^2 U^T$ and $\tilde{x}_i = U_K U_K^T x_i$

Relationship between eigenvalues:

$$\Lambda = \frac{1}{N} \Psi^2 \quad \Rightarrow \quad \text{PCA}(X) \leftrightarrow \text{AE}\left(\sqrt{\frac{1}{N}} X\right)$$

Fundamental Conclusion: A linear autoencoder performs PCA if the data is **centered and normalized**

Note: W_{enc} is **not the inverse** of W_{dec} if $K < D$

Extensions of Autoencoders

Nonlinear Autoencoders:

By adding hidden layers with nonlinear activation functions, the latent space becomes **nonlinear**.

Variational Autoencoders (VAE):

By changing the loss function (using ELBO - Evidence Lower BOund), we can impose **sparsity** in the latent space.

Alternative Formulations of PCA

Probabilistic PCA

Reformulation with Explicit Latent Distribution:

- Latent distribution: $\mathbb{P}(z) = \mathcal{N}(0, I_{dK})$
- Conditional model: $\mathbb{P}(x|z) = \mathcal{N}(Wz + \mu, \sigma^2 I_{dD})$

Advantages:

- Accommodation of **measurement noise** (variance σ^2)
- **Generative mode**: possibility to generate new data
- Parameter estimation by **Maximum Likelihood**
- **EM algorithm** (Expectation-Maximization) to save computations
- **Bayesian PCA**: inversion of the conditional to find K by training

Kernel PCA

Principle:

Use a **nonlinear mapping** $\phi(x_i)$ via a kernel function:

$$k(x_i, x_j) = \phi(x_i)^T \phi(x_j)$$

Allows performing PCA in a **more favorable space** (possibly infinite-dimensional) without explicitly computing ϕ.

Local PCA

Performs PCA on **local neighborhoods** of the data to consider only the **local intrinsic dimensionality**.

Useful for data with complex local structure (nonlinear manifolds).

Computational Limitation

Complexity: The main limitation of classical PCA is the **eigendecomposition** of Σ in $O(D^3)$

For very large dimensions, alternative or approximate methods may be necessary.

Synthesis and Key Points

Essential Characteristics of PCA

Nature of the Model:

- PCA is part of **linear latent models**
- Applies to **centered data**
- Uses **variance** as a decomposition criterion

Mathematical Properties:

- **Exact and complete** decomposition into **decorrelated** components
- Assumes a **normal distribution** of data
- Can be formulated as a **regression**

Applications:

- **Justified decimation** strategy based on low-rank approximation
- Useful for **denoising** via a Gaussian noise model

Equivalences:

- Equivalent to a **linear autoencoder**
- Can receive a **stochastic formulation**
- Generalizable to the **nonlinear case via kernels**

When to Use PCA?

Suitable for:

- Data approximately following a Gaussian distribution
- Dimensionality reduction with variance preservation
- Visualization of high-dimensional data
- Denoising data corrupted by Gaussian noise
- Data compression

Not Suitable for:

- Strongly clustered data (non-Gaussian)
- Data with strongly nonlinear relationships
- Cases where variance is not a good information criterion

Mathematical Concepts to Master

To understand PCA well, you need to master:

- **Linear transformation** and **orthogonal matrix**
- **Coordinates** and **matrix rank**
- **Mean** and **variance**
- **Orthogonal projection**
- **Eigendecomposition**
- **Matrix trace**
- **Frobenius norm**
- **Lagrange multipliers**

Typical Exam Questions

Conceptual Questions

1. Explain how PCA uses variance as a criterion.
2. Show that maximizing variance is equivalent to minimizing error.
3. Show how PCA uses the Eckart-Young theorem.
4. Can PCA be applied to any data?
5. Is it relevant to apply PCA to any data?

6. How to apply PCA to clustered data?

Technical Questions

Given data:

- How do you apply PCA?
- What information does PCA provide?
- How to reconstruct data with $K < D$ components?
- How to select the components to retain?

Mathematical Proofs

- Show that PCA is equivalent to a linear autoencoder.
- Demonstrate the equivalence between maximum variance and minimum error.
- Prove that principal components are the eigenvectors of Σ.